

RONNIE KIYEGGA | Full Stack Software Engineer

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Full-stack software engineer with 4+ years of experience building and shipping web platforms (React/Next.js, Node.js, PostgreSQL) across EdTech, e-commerce, and B2B SaaS. Delivers products end-to-end from design prototyping to CI/CD-deployed production with a track record of reducing delivery cycles, cutting infrastructure costs, and building real-time analytics systems serving cross-functional teams.

SKILLS

Languages: TypeScript, Python, SQL
Frontend: React, Next.js, Tailwind CSS, HTML5, CSS3
Backend: Node.js, Express.js, RESTful APIs, PostgreSQL, MongoDB, Prisma, TensorFlow.js
Testing: Jest, Cypress, React Testing Library
Infrastructure: Docker, Nginx, GitHub Actions, DigitalOcean, Vercel, AWS S3

EXPERIENCE

Software Engineer & CS Teacher

Aug 2023 – Jan 2026

School Of Research Science | Dubai, UAE

- Led end-to-end architecture and delivery of a browser-based pseudocode IDE (Next.js, TypeScript, PostgreSQL) with a real-time semantic analysis engine, exam-board-aligned AST validation, and Stripe-integrated subscription management, scaling to 1000+ active users across including a live GCSE and A-Level cohort.
- Owned the design and delivery of a client-side KNN image classification pipeline (TensorFlow.js) optimised for memory-constrained iPad hardware replacing an external vendor dependency (Google Teachable Machine) and enabling on-device ML inference without server round-trips.
- Led design and delivery of a multi-department real-time analytics platform (React, Chart.js) serving 4 teams, cutting end-of-term report generation from 20+ hours of manual processing to under 1 hour per cycle.
- Drove QA cycle time from 2 days to 4 hours across 3 production apps by owning the test infrastructure implementing automated linting, Jest unit tests, and Cypress E2E tests, enabling same-day releases.

Freelance

Jun 2020 – Aug 2023

Full-Stack Software Engineer | London, UK

- Engineered and maintained a 30+ component design system (React, TypeScript) reused across 8+ e-commerce and media client projects, cutting per-client delivery time by 33% (6 weeks → 4 weeks) from Figma prototype to production.
- Owned end-to-end delivery of 8+ full-stack client projects (Next.js, Stripe, Sanity CMS) for D2C e-commerce brands and media companies from infrastructure provisioning on Vercel/AWS through to stakeholder handoff with zero critical incidents over a 3-year engagement.
- Spearheaded a 40% reduction in media infrastructure costs by replacing third-party video hosting with a custom Node.js/FFmpeg transcoding pipeline on AWS S3, generating optimised 4K/HD/mobile formats served via pre-signed URLs.

Adaptive - London, UK

Aug 2019 – Dec 2019

Software Engineer Internship

- Built and shipped self-service React dashboards with D3.js timeline visualisations for 3 fintech product teams, cutting analyst query turnaround from 2–3 days to minutes; resolved UI performance bottlenecks in the shared component library by optimising memoisation patterns.

PROJECTS

EduFeedbackPro (B2B SaaS AI-Analytics Platform)

github.com/BlissfulCoda/edufeedbackpro

- Architected a 3-tier data pipeline (BigQuery → NeonDB → MongoDB) with NeonDB as a serverless cache layer, achieving sub-50ms P95 API response times and cutting BigQuery query costs for analytics serving 1,200+ students.
- Migrated from WebSockets to SSE with Sentry alerting, reducing issue detection time from 30+ min to under 2 min; integrated ElevenLabs voice AI for hands-free at-risk student queries.

Ms Maryam's Math (B2C Tutoring SaaS — In Progress)

msmaryamsmaths.com

- Owned full-stack build with JWT-based RBAC (3 roles), Arcjet rate limiting, and blue-green CI/CD via GitHub Actions — reducing deployment time from 20 min to 3 min with zero downtime.

EDUCATION

PGCE Computer Science • King's College London (*Postgraduate teaching qualification*)

Sep 2021 – Jun 2022

BSc (Hons) Computer Science • King's College London

Sep 2017 – Jun 2020